

Answer Key — General Science Grade 10

Sub-Strand 1.3: Nutrition in Animals • Lesson 7: Model Building — Making a 3D Model of the Human Digestive System

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	C	B	A	D	D	C

USE AFTER: PREDICT PHASE — Predict Phase: individual sketch of the alimentary canal from memory

Q1. A learner sketches the journey of a piece of ugali through the alimentary canal. Which sequence of organs is correct?

- A. Mouth → oesophagus → stomach → large intestine → small intestine → rectum
- B. Mouth → oesophagus → small intestine → stomach → large intestine → rectum
- C. Mouth → oesophagus → stomach → small intestine → large intestine → rectum**
- D. Mouth → stomach → oesophagus → small intestine → large intestine → rectum

Correct answer: C. Food passes from the mouth down the oesophagus to the stomach, then the small intestine, the large intestine and finally the rectum. The other options swap organs out of order.

USE AFTER: OBSERVE PHASE — Observe Phase: Evidence Retrieval Gallery and building list (organ, process, enzyme + substrate)

Q2. A group is writing its building list for the mouth. Which entry is correct for the enzyme and substrate in the mouth?

- A. Salivary amylase acts on proteins to form peptides
- B. Lipase acts on fats to form fatty acids and glycerol
- C. Salivary amylase acts on starch (as in ugali) to form maltose**
- D. Pepsin acts on proteins to form peptides

Correct answer: C. Salivary glands release amylase, which begins the chemical digestion of starch into maltose. Pepsin works in the stomach and lipase acts mainly in the small intestine.

USE AFTER: EXPLAIN PHASE — Explain Phase: model construction, teacher probe on liver/bile entering the small intestine

Q3. A group is building its model and is asked: what does bile do to the fat in nyama choma, and why is that important?

- A. Bile carries digested fat across the villi into the bloodstream
- B. Bile emulsifies fat into tiny droplets, giving lipase a larger surface to act on**
- C. Bile is an enzyme that chemically splits fat into fatty acids, so lipase is not needed
- D. Bile is made in the stomach and digests the proteins in the meat

Correct answer: B. Bile is made in the liver and stored in the gall bladder. It contains no enzymes, but it breaks large fat globules into small droplets so lipase can digest the fat efficiently.

USE AFTER: EXPLAIN PHASE — Explain Phase: model construction, teacher probe on the large intestine with no enzyme labels

Q4. A group labels the large intestine on its model. Which label is correct?

- A. Water is reabsorbed here; no digestive enzymes are made here**
- B. Pepsin digests proteins into peptides
- C. Maltase and sucrase finish the digestion of sugars
- D. Lipase digests the remaining fats

Correct answer: A. The large intestine mainly reabsorbs water from undigested food. It has no digestive enzymes, so a label without an enzyme is correct, not a mistake.

USE AFTER: DQB PHASE — DQB Phase: selecting question cards the model answers, linking to the 3 cm tube puzzle

Q5. Damp sponge strips are placed inside the tube that represents the small intestine. What do these strips represent, and why do they matter?

- A. Bile, which breaks up fat into small droplets
- B. Sphincters, which stop food from moving backwards
- C. Stomach walls, which churn the food mechanically
- D. Villi, which greatly increase the surface area for absorbing nutrients into the blood**

Correct answer: D. The sponge strips model villi, the finger-like folds of the small intestine. They give a very large surface area, which explains how a tube only about 3 cm wide can absorb nutrients from a full school lunch.

USE AFTER: MODEL BUILDING PHASE — Model Phase: whole-class question, "where is the idea that the stomach digests everything wrong?"

Q6. At the start of the unit, a learner said, "The stomach digests everything in our ugali, sukuma wiki and nyama choma." Which statement best shows where this idea is wrong?

- A. The stomach digests starch fully, so the small intestine only absorbs water
- B. The stomach only stores food, and no digestion happens there
- C. The stomach makes bile, so it digests fats but not proteins
- D. The stomach starts protein digestion with pepsin and HCl, but most chemical digestion and nearly all nutrient absorption happen in the small intestine**

Correct answer: D. In the stomach, churning, pepsin and HCl begin protein digestion only. Fats, starch and peptides are digested further in the small intestine, where villi absorb the nutrients. Bile is made in the liver, not the stomach.

USE AFTER: END OF LESSON — Model Phase: 'Failure Mode' tag extension for groups that finish early

Q7. A group adds a red 'Failure Mode' tag to the pancreas on its model. Which tag correctly describes what happens if the pancreas stops producing lipase?

- A. Starch in the ugali cannot be digested in the mouth
- B. Proteins in the meat cannot be digested in the stomach
- C. The fat in nyama choma is not digested and passes into the large intestine unabsorbed**
- D. The large intestine cannot reabsorb water from the waste

Correct answer: C. Lipase digests fats, and the pancreas supplies it to the small intestine. Without lipase, fat stays undigested and moves on unabsorbed. Starch digestion in the mouth, protein digestion in the stomach and water reabsorption do not depend on pancreatic lipase.